INCH-POUND

MIL-S-63433D(AR)
20 January 1989
SUPERSEDING
MIL-S-63433C(AR)
24 April 1987

MILITARY SPECIFICATION

SIMULATOR, ARTILLERY FLASH, M21, PERFORMANCE SPECIFICATION FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for design and establishes a uniform method of testing one type of Artillery Flash Simulator, M21. The purpose of testing is to determine performance, safety, soundness of design and resistance to environments encountered during storage, handling, and service use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of.
MIL-A-2550	- Ammunition and Special Weapons; General Specification For.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9342970	- Packing & Marking for Box, Paperboard, for Simulator, Artillery Flash, M21.
9342972	- Packing & Marking for Box, Packing, for Simulator, Artillery Flash, M21.

TEST PROCEDURES

TOP 1-2-511	- Electromagnetic Radiation Effects and Hazards Test.
TB 700-2	- DoD Explosive Hazard Classification Procedures.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

CODE OF FEDERAL REGULATIONS

TITLE 49 Transportation, Parts 100-199

(The Code of Federal Regulations is available for the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199. (latest revision))

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-117 - Salt Spray (Fog) Testing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Assembly. The simulator assembly shall correctly interface with the gunfire device manufactured by Hoffmann-Werke designated as Simulator, Tank, Gunfire Device 17-61, EC Corporation device designated as Simulator, Tank, Gunfire Device 17-157, and the Gunfire Simulator, Device 17-133 manufactured by Joanell Laboratories, and shall comply with all requirements specified in applicable specifications and standards. (ref. 6.5.3)

3.2 Materials.

3.2.1 Nonmetals. Nonmetal materials such as sealants, adhesives and plastics shall not deteriorate with age or change properties to the extent that the simulators fail to conform to the functional requirements specified under 3.4 of this specification at any time during the warranty period, or five years whichever is longer when packaged as specified in 5.4.

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3.2.2 Metals. Metals shall be corrosion-resistant or shall be suitably treated to resist corrosion due to salt spray when tested as specified in 4.5.7.

3.2.3 Dissimilar metals. Dissimilar metals shall not be used in contact with each other, unless suitably protected against electrolytic corrosion. Dissimilar metals are defined in MIL-STD-889.

3.2.4 Flash charge sensitivity. The contractor shall provide sensitivity data on all energetic materials unique to the system. The sensitivity test documents and procedure descriptions are as follows (for unique material):

3.2.4.1 Impact. The composition shall be subjected to the impact test as specified in 40.1 of Appendix A to this specification. The 50% point shall not be less than 20 centimeters.

3.2.4.2 Friction pendulum. The composition shall be subjected to the friction pendulum test as specified in 40.2 of Appendix A to this specification. The energetic material may burn, but may not detonate when using the steel shoe.

3.2.4.3 Electrostatic discharge. The composition shall be subjected to the electrostatic discharge test as specified in 40.3 of Appendix A to this specification. There shall be 20 out of 20 no-fires at 0.25 joules.

3.2.4.4 Explosive temperature. The composition shall be subjected to the explosive temperature test as specified in 40.4 of Appendix A to this specification. The explosive temperature shall be 200 degrees centigrade (^{o}C) minimum (min).

3.2.4.5 Vacuum thermal compatibility. The composition shall be subjected to the vacuum thermal compatability test as specified in 40.5 of Appendix A to this specification. There shall be not more than 2 milliliters of gas evolved per 5 grams of energetic material.

3.2.4.6 Auto-ignition temperature. The composition shall be subjected to the auto-ignition temperature test as specified in 40.6 of Appendix A to this specification. The auto-ignition temperature shall be 200°C min.

3.2.5 Health and safety. This specification contains requirements which may necessitate the handling of hazardous materials and result in the manufacture of a hazardous item. It is emphasized that all applicable safety rules, procedures, and regulatory requirements be strictly followed during the handling, processing, and testing of the completed item and related hazardous materials. All required safety information shall be submitted to the Contracting Officer for approval.

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3.2.5.1 Simulator explosive hazard classification. A copy of the DOT authorization shall be furnished to the Contracting Officer prior to delivery of any assemblies, items, or hazardous materials. Transportation of the item must meet the requirements of the CODE OF FEDERAL REGULATIONS TITLE 49 - TRANSPORTATION, PARTS 100-199. The simulator, in its shipping and storage packaging, shall meet the requirements of the following classifications when tested in accordance with 5.3.f "Stack Test" and 5.3.g "External Fire Stack Test" of TB 700-2:

DoD Hazard Classification - 1.3
DoD Compatability Group - G
DoT Hazard Classification - Class B Explosives
DoT Marking - SPECIAL FIREWORKS - HANDLE CAREFULLY - KEEP FIRE AWAY

Prior to the acceptance of the simulator by the Government, the Contracting Officer shall review the contractor's baseline Tech Data Package to determine if this classification can be obtained by analogy or if testing in accordance with TB 700-2 is required.

3.2.5.2 DOT authorization/hazard classification. Certification from the supplier shall be required to assure no significant change in the item, materials, packaging, etc. which would affect the DOT authorization/hazard classification.

3.2.5.3 Hazardous materials (energetic, toxic, carcinogenic). Hazardous materials/components will be identified as they are selected, and submitted to the Contracting Officer for approval. Only energetic materials acceptable for DOD use will be utilized.

3.2.5.4 Explosive ordnance disposal (EOD). EOD render safe procedures and applicable render safe procedure tools will be formulated as early as possible and will be kept current with design. These procedures and tools shall be made available to the Government for approval as early as possible to insure applicable tools, equipment and publications will be available to army EOD units 30 days before the deployment date of the simulator.

3.2.5.5 Demilitarization. Demilitarization/disposal procedures will be formulated and kept current with design changes as they are instituted. These procedures shall be made available to the Government for approval at the time of submission of the first article sample.

3.2.5.6 Static and other induced electric hazards. The ignition system should be designed and maintained so that static charges and induced electric energies will not produce a hazardous device.

3.2.5.6.1 Electromagnetic radiation susceptibility. The simulator shall not function when tested as specified in 4.5.14.

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3.2.5.6.2 Electrostatic discharge (ESD). The simulator shall not function from human-borne or aircraft-borne ESD when tested as specified in 4.5.15.

3.3 Design and construction. The components of the simulator shall be designed and constructed to conform to the performance requirements specified herein.

3.3.1 Assembly of components. All components shall be made from either plastic, paper or both products with exception of the connecting wires, contacts and pyrotechnic compositions. The components of the simulator shall consist of at least a container, flash charge, igniter, connecting wires and plug assembly and some method of protecting the item from inadvertant or accidental initiation.

3.3.2 Application of sealants and adhesives. Any sealants or adhesives used in the construction of the simulator shall be applied in such a manner that the performance of the simulator in 3.4 after prolonged storage shall not be degraded by aging or any other changes of the sealant or adhesive. A sealant used to seal the interface of a cap and container shall be applied in such a manner that excessive sealant will not enter the interior of the simulator and cover or encapsulate the chemical components in such a way as to change the functional characteristics of the simulator under prolonged storage.

3.3.3 Plastic parts. Plastic components shall be compatible with the flash charge and shall not degrade during prolonged storage to the extent that the simulators fail to meet the functioning requirements of 3.4.

3.3.4 Moisture content.

3.3.4.1 Charge composition. The moisture content of the charge composition at the loading station at the time of loading shall not exceed 0.10 percent when tested as specified in 4.5.3.1.

3.3.4.2 Paper and paper products. The moisture content of any paper or paper product component of the simulator assembly at the loading station at the time of loading shall not exceed 6.0 percent when tested as specified in 4.5.3.2.

3.3.5 Bridge-wire resistance. The igniter bridge-wire resistance shall be between 1.1 and 1.4 ohms when tested as specified in 4.5.9.

3.3.6 Connecting wire assembly resistance. The connecting wires shall be 24.6 + .4 inches in length fitted with a plug that interfaces with the applicable gunfire devices. The assembly resistance, including connecting wires, bridgewire and plug shall be 1.51 ohms max. when tested as specified in 4.5.13.

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3.3.7 Shorting cap. The plug shall be fitted with a shorting cap for safety purposes.

3.3.8 Label. The label shall be as shown in Figure 1.

3.4 Performance. The simulator shall not damage the gunfire device or interface with repeated use, and shall satisfy all requirements specified herein after subjection to the applicable environmental tests. Performance shall be measured and values recorded in accordance with 6.4 for the ballistic operation of the simulator.

3.4.1 Functioning. The assembly shall function and comply with all requirements when tested in accordance with 4.5.4 after conditioning at the specified temperatures.

3.4.1.1 Sound characteristics. The sound level intensity of the simulator shall be a minimum of 124 decibels (db) to a maximum (max.) of 147 db when measured at a distance of 96 meters min. from the item under test when tested as specified in 4.5.4.1.

3.4.1.2 Functioning time. The time from application of 24 + 2 Volts Direct Current (VDC), with a minimum ignition energy of 3.0 milliwatt seconds/ohm, to initiation of the assembly shall not be greater than .50 seconds (sec) when tested as specified in 4.5.4.2.

3.4.1.3 Flash characteristics. The flash intensity of the simulator shall be 4 x 10^6 candlepower minimum when tested as specified in 4.5.4.3. Residual smoke shall be white in appearance.

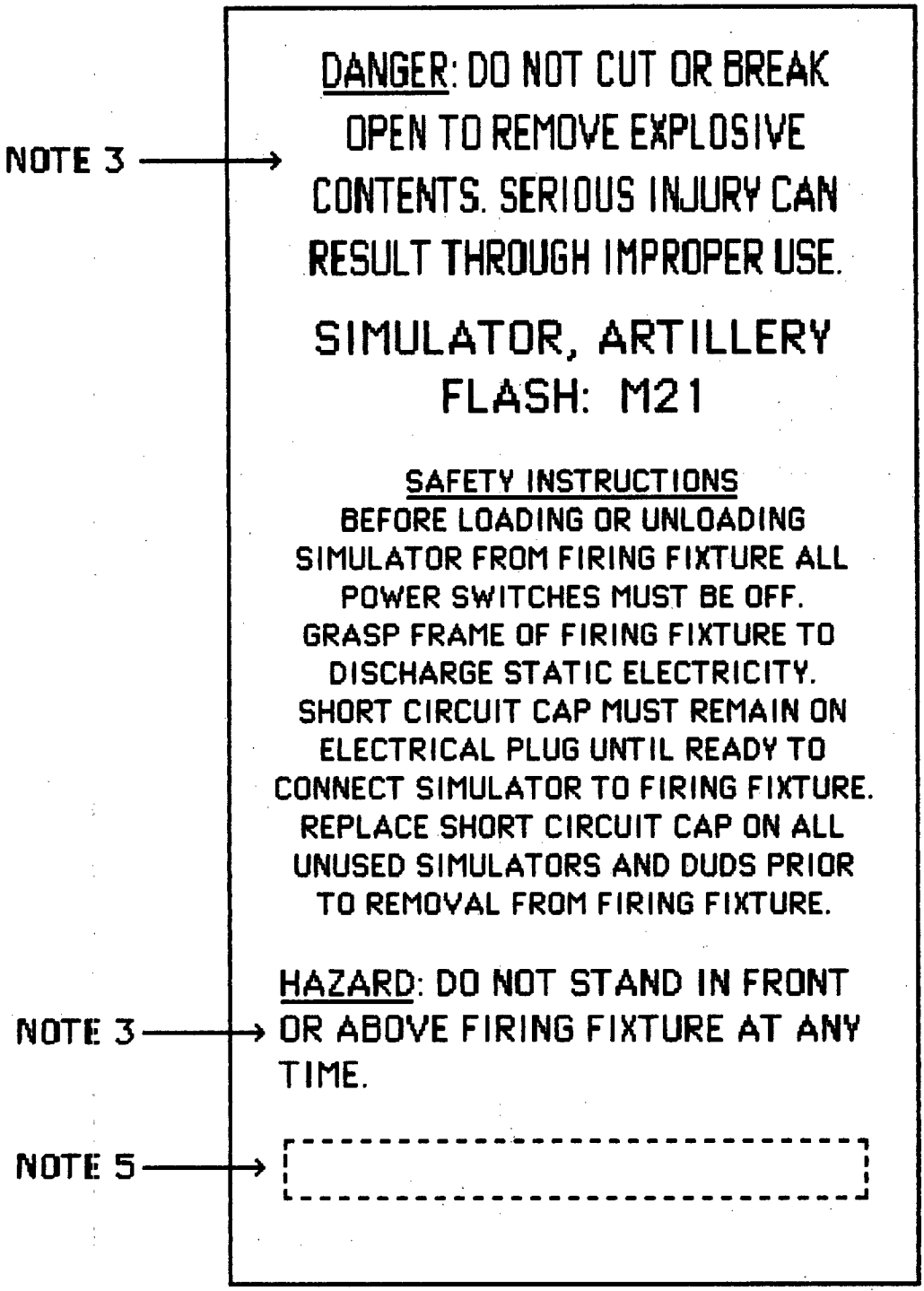
3.4.2 Chamber withdrawal. The simulators shall be capable of being extracted freely from the firing tubes of approved gunfire devices when a force of three (3) to twelve (12) pounds is applied at a temperature of 70±5°F when tested as specified in 4.5.2.

3.4.3 Functional timing sequencing of the electric igniter. The continuity of the electric igniter shall be 10 milliseconds (ms) minimum at 12 VDC (nominal) at 0.5 amperes (amp). It shall produce an open circuit (if only for an instant) within 25 ms of receiving 12 VDC at 1 ampere when tested as specified in 4.5.10.

3.4.4 Surface danger zone. When fired, the simulator shall produce a hazardous fragment distance of no greater than 25 meters to either side or 50 meters forward of the firing device when tested as specified in 4.5.12.

3.4.5 Performance reliability. The simulator must meet all performance requirements at a reliability of not less than 95 percent at a confidence level of 90 percent over a temperature range of 125 ± 5°F to minus 25 ± 5°F as defined in 4.3.3 and 4.4.2.

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NOTES:

1. PRINTED INFORMATION SHALL BE WATER AND SMUDGE RESISTANT.
2. ADHESIVE SHALL PASS WATER IMMERSION TEST.
3. PRINT WITH RED INK.
4. PRINT REMAINDER WITH BLACK INK.
5. INSERT LOT NUMBER.

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Figure 1

3.5 Environmental conditions.

3.5.1 Vibration. The simulators, when packed, shall not fire when subjected to vibration as specified in 4.5.8.1. After vibration (and additional testing as indicated for first article tests), the simulators shall show no signs of damage, shall be safe to handle and operate, and shall meet the performance requirements of 3.4.1.

3.5.2 Five foot drop. The bare simulator shall not fire when dropped from a height of 5 feet as specified in 4.5.8.12. After drop, the simulator shall be safe to handle and operate, and shall meet the applicable requirements of 3.4.1.

3.5.3 Seven-foot drop. The simulators, when packed, shall not fire when subjected to 7 foot drop as specified in 4.5.8.6. After being subjected to drop test (and additional testing as indicated for first article tests), the simulators shall show no signs of damage, shall be safe to handle and operate, and shall meet the performance requirements of 3.4.1.

3.5.4 Loose cargo. The simulators, when packed, shall not fire when subjected to loose cargo testing as specified in 4.5.8.8. After loose cargo test (and additional testing as indicated for first article tests and quality conformance inspection), the simulators shall show no signs of damage, shall be safe to handle and operate, and shall meet the performance requirements of 3.4.1.

3.5.5 Ten foot drop. The bare simulator shall not fire when dropped from a height of 10 feet as specified in 4.5.8.5. After being subjected to drop, the simulators, if not damaged to the extent that they cannot be loaded into the gunfire device shall be safe to operate. If after such a drop, the simulators are damaged to the extent that they cannot be loaded into the gunfire device, they shall be safe to handle and dispose of.

3.5.6 Forty-foot drop. The simulators, when packed, shall not detonate when dropped from a height of 40 feet as specified in 4.5.8.7 and shall be safe to handle and dispose of.

3.5.7 Cold temperature. The bare simulator shall be capable of meeting the performance requirements of 3.4.1 after being subjected to cold temperature conditions as specified in 4.5.8.2.

3.5.8 Hot temperature. The bare simulator shall be capable of meeting the performance requirements of 3.4.1 after being subjected to hot temperature conditions as specified in 4.5.8.3.

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3.5.9 Leakage test. The simulators, when packaged, shall be capable of meeting the performance requirements of 3.4.1 after exposure to the leakage test as specified in 4.5.6.

3.5.10 Water immersion test. The bare simulator shall meet the performance requirements of 3.4.1 after exposure to water immersion as specified in 4.5.8.11.

3.5.11 Jumble. The bare simulator shall comply with the following requirements when tested as specified in 4.5.5.

a. The simulator shall not function.

b. There shall be no powder leakage.

c. Components shall not separate.

3.5.12 Temperature-humidity. The packed simulators shall meet the performance requirements of 3.4.1 after exposure to temperature-humidity cycle as specified in 4.5.8.9.

3.5.13 Temperature shock. The bare simulator shall meet the performance requirements of 3.4.1 after being subjected to the temperature shock as specified in 4.5.8.10.

3.6 First article inspection. This specification provides for the conduct of first article inspection (qualification test) to establish that the simulator meets the aforementioned requirements. Successful completion of this test establishes a design baseline which defines the designs, materials and production processes by which the simulator is produced. The contractor shall then submit simulator drawings in sufficient detail to establish the baseline design (ref. 6.5). No change to that baseline may be made without the approval of the Procuring Contracting Officer (PCO). The PCO may require the submission of a complete or partial first article test sample prior to permitting any change to the established baseline.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which could affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any or all of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.2 and 3.3. If material defects, incompatability, or inferior quality is evident, and the Government deems material analysis necessary, the contractor may be requested to submit samples of specimens to the appropriate design agency for analysis and approval.

4.2 Classification of tests. The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of Table I.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in this specification.

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4.3.3 Rejection.

a. Each of the environmental test samples shall pass the accept/reject criteria for each specified environment.

b. The entire sample tested (12 functioning tests for a total of 918 samples) shall pass an accept/reject criteria of 918-30-31.

c. A simulator shall be classified as a functional failure if it fails any of the requirements of 3.4.1.

d. Any simulator, upon functioning, that damages the gunfire device making any tube unuseable shall reject the First Article.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63433D(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 1 OF 4		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Bridge-wire resistance	20	0-1	3.3.5	4.5.9
	Connecting wire assembly resistance	20	0-1	3.3.6	4.5.13
	Functional timing sequencing test	20	0-1	3.4.3	4.5.10
	Chamber withdrawal test	20	0-1	3.4.2	4.5.2
	Moisture content				
	Charge composition	25	0-1	3.3.4.1	4.5.3.1
	Paper or paper products	25	0-1	3.3.4.2	4.5.3.2
	Salt spray test	20	0-1	3.2.2	4.5.7
	Electromagnetic radiation	1 box	0-1	3.2.5.6.1	4.5.4.4
	Electrostatic Discharge	1 box	0-1	3.2.5.6.2	4.5.4.5

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433D(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 2 OF 4		DRAWING NUMBER / NEXT HIGHER ASSEMBLY	
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD	
			Ac/Re			
	Surface Danger Zone	9	0-1	3.4.4	4.5.12	
	Ten Foot Drop	30	0-1	3.5.5	4.5.8.5	
	Non-sequential test					
	Temperature cycling			3.5.13	4.5.8.10/4.5.4	
	Cold, Ambient	30	1-2			
	Cold, Ambient, Hot	30	1-2			
	Cold, Hot	30	1-2			
	Cold, Hot, Ambient	30	1-2			
	Hot, Ambient	30	1-2			
	Hot, Cold	30	1-2			
	Jumble - Ambient	30	0-1	3.5.11	4.5.5	
	Forty foot drop			3.5.6	4.5.8.7	
	Cold	162 (1 box)	0-1			
	Hot	162 (1 box)	0-1			
	Temperature-humidity	162 (1 box)		3.5.12	4.5.8.9	
	Ambient Temperature (Note 1)	30	1-2	3.4.1	4.5.4	
	Cold Temperature (Note 2)	30	1-2	3.5.7	4.5.8.2/4.5.4	
	Hot Temperature (Note 2)	30	1-2	3.5.8	4.5.8.3/4.5.4	

NOTES: 1. Simulators shall be conditioned for a minimum of 6 hours at 70 ± 5°F after undergoing temperature Humidity Test. 2. Simulators to be functioned shall first undergo temperature-humidity test & applicable temperature conditioning. (Hot 125±5°F, Cold -25±5°F)

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433D(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 3 OF 4		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Heat seal of barrier bag	18 (1 box)		5.4	4.5.11
	Sequential Test (NOTE 6)				
	Leakage test (prior to shipment) (NOTE 3)	648 (4 boxes)		3.5.9	4.5.6
	Vibration			3.5.1	4.5.8.1
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Seven foot drop			3.5.3	4.5.8.6
	Cold	324 (2 boxes)	0-1		

NOTES:
3. Leakage test performed on barrier bags then repacked into boxes for sequential testing.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433D(AR)

PARAGRAPH CATEGORY	TITLE Simulator, Flash, Artillery, M21	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	SHEET 4 OF 4 DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Hot	324 (2 boxes)	0-1		
	Loose cargo			3.5.4	4.5.8.8
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Water immersion test	648	15-16	3.5.10	4.5.8.11
	Five foot drop				
	Safety	60	0-1	3.5.2	4.5.8.4
	Cold (Note 4) - function	30	1-2	3.4.1	4.5.4
	Hot (Note 4) - function	30	1-2	3.4.1	4.5.4
	Functioning-Ambient (Note 5)	588	20-21	3.4.1	4.5.4

NOTES: 4. Fifteen (15) of these items shall be taken from each of the respective boxes. These items shall be subjected to water immersion prior to functioning. 5. Simulators shall be conditioned at ambient ($70\pm5^{o}F$) temperature for a minimum of 6 hours prior to functioning. 6. All 0-1 criteria (on Sequential Tests only) pertain to premature function, damage, and safe to handle after testing and prior to ambient function.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. See MIL-A-48078.

4.4.2 Examination. The examinations/tests listed herein shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the classification of defects or test tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Level II.

b. Each of the environmental test samples shall pass the accept/reject criteria for each specified environment.

c. The entire sample tested (three functioning tests for a total of 125) shall pass an accept/reject criteria of 125-4-5.

d. A simulator shall be classified as a functional failure if it fails any of the requirements of 3.4.1.

e. Any simulator, upon functioning, that damages the gunfire device making any tube unuseable shall reject the lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63433D(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Simulator, Flash, Artillery, M21	1	2	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Moisture content				
	(Major defect)				
	Charge composition	25	0-1	3.3.4.1	4.5.3.1
	Paper products	25	0-1	3.3.4.2	4.5.3.2
	Bridgewire resistance test	25	0-1	3.3.5	4.5.9
	(Major defect)				
	Chamber withdrawal test				
	(Major defect)	25	0-1	3.4.2	4.5.2
	Function test (Ambient) NOTE 1	50	2-3	3.4.1	4.5.4
	Sound characteristics		NOTE 2	3.4.1.1	4.5.4.1
	Flash characteristics			3.4.1.3	4.5.4.3

NOTES: 1. Cold (minus 25 + 5°F) Ambient (70 + 5°F) Hot (125 + 5°F), to be conditioned for six hours (min) and fired within five minutes after removal from conditioning chamber.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63433D(AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1 (cont'd)	Simulator, Flash, Artillery, M21		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Function test (Hot)	50	2-3	3.4.1	4.5.4
	Sound characteristic		NOTE 2	3.4.1.1	4.5.4.1
	Flash characteristic			3.4.1.3	4.5.4.3
	Function test (Cold)	25	2-3	3.4.1	4.5.4
	Sound characteristic		NOTE 2	3.4.1.1	4.5.4.1
	Flash characteristic			3.4.1.3	4.5.4.3

NOTES:
2. No more than four (4) defects shall be allowed for the total 125 function test sample.

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables.

4.4.3.1 Moisture content (see Table II) - Major Defects.

TABLE II - Moisture Content

Charge composition	(see 3.3.4.1)
Paper and paper products	(see 3.3.4.2)

The Contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. If a composite sample exceeds the requirement, that quantity or sub-lot material represented by the sample shall not be used in loading or packing. The remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Table I and 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Examination of product. Examination of the simulators shall be made to detect visible defects and to determine that the items meet the requirements of this specification.

4.5.2 Chamber withdrawal test. The simulators shall be installed in the approved gunfire device by pushing down until top of simulator is flush with end of firing tube. The approved gunfire tube shall be clean and dry. Slowly apply a pull force on the simulator using an approved method in accordance with 4.4.4. Observation shall be made for compliance with the applicable requirements. (Non-Destructive test).

4.5.3 Moisture content.

4.5.3.1 Charge composition. The method of determining the percentage of moisture in the flash charge shall be in accordance with Test Method 102.2.1 of MIL-STD-1234 except that the sample shall be heated until a constant weight is obtained in lieu of 4 hours. Observation shall be made for compliance with the applicable requirements.

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4.5.3.2 Paper or paper products. The moisture content shall be determined in accordance with Test Method 102.1.1 of MIL-STD-1234 except that a 10 gram sample shall be used and the drying time shall be for 2 hours. Observations shall be made for compliance with the applicable requirements.

4.5.4 Functioning. The simulator shall be conditioned for 6 hours min. at the specified temperatures and fired within 5 minutes after removal from the conditioning chamber. The simulator shall be mounted horizontally in an approved gunfire device and fired by means of an approved power source in accordance with 3.4.1.2. (Destructive test)

4.5.4.1 Sound characteristics. The sound level intensity shall be measured using approved metering equipment in accordance with 4.4.4. The simulator shall be suspended 4.9 $\pm$ 0.2 meters off the ground. The detector shall be positioned 1.5 $\pm$ 0.1 meters off the ground at a side position to the longitudinal axis of the simulator. Observations shall be made for compliance with applicable requirements.

4.5.4.2 Functioning time. The time from the application of the 24 VDC to the item functioning shall be measured with approved metering equipment in accordance with 4.4.4. The simulator shall be observed for compliance with the applicable requirements.

4.5.4.3 Flash characteristics. The flash intensity of the simulator shall be recorded using an approved method in accordance with 4.4.4. The simulator and detector positioning shall be the same as 4.5.4.1. Observations shall be made for compliance with the applicable requirements.

4.5.5 Jumble. The simulator shall be tested in accordance with procedures specified in MIL-STD-331, Test 102, except that the duration of the test shall be a minimum of 20 minutes. The simulator and equipment shall be observed and inspected for compliance with the applicable requirements. (Destructive test).

4.5.6 Leakage test. The test shall be performed in accordance with the procedure specified in 4.4.3.1 and 4.4.3.2 or 4.4.3.6 of MIL-P-116. After the test, the seal shall be examined for compliance with the applicable requirements. (Non-Destructive test)

4.5.7 Salt spray test. The metal parts shall be subjected to the salt spray test in accordance with ASTM-B-117 using a 5% atomized sodium chloride solution. Exposure time shall be not less than 2 hours. Examination shall be made for any evidence of corrosion or salt deposits indicative of improper coating. Any evidence of improper coating shall cause the part to be classed defective. (Destructive test)

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4.5.8 Environmental tests. Environmental testing shall be conducted to determine compliance with the requirements of 3.5. The packed configuration used in the subsequent tests shall be in accordance with 5.5.

4.5.8.1 Vibration. This test shall be conducted in accordance with Method 514.3 MIL-STD-810, Test Procedure I, Test Condition I-3.2.1, Figures 514.3-1, 514.3-2, 514.3-3 (Vertical, Transverse and Longitudinal axis) for 120 minutes (equal to 2000 miles) in each axis. Observations shall be made for compliance with the applicable requirement. (Non-Destructive test)

4.5.8.2 Cold temperature. The simulators shall be subjected to a temperature of minus 25 ± 5°F until thermal stabilization is reached and maintained for a period of 6 hours minimum. The test specimen shall be removed from the conditioning cabinet, set up, and fired within a period of time not to exceed 5 minutes. Observations shall be made for compliance with the applicable requirement. (Destructive test)

4.5.8.3 Hot temperature. The simulators shall be subjected to a temperature of 125 ± 5° until thermal stabilization is reached and maintained for a period of 6 hours min. The test specimen shall be removed from the conditioning cabinet, set up, and fired within a period of time not to exceed 5 minutes. Observations shall be made for compliance with the applicable requirement. (Destructive test)

4.5.8.4 Five foot drop test. The simulators shall be dropped from a height of 5 feet onto a 3 inch thick steel plate embedded in concrete. Thirty (30) items will be tested at hot temperature and thirty (30) items tested at cold temperature for First Article Testing. Six (6) items in each temperature phase will be dropped for each of the five orientations listed below. The simulators will be placed in an approved gunfire device and functioned following the drop. Observation shall be made for compliance with the applicable requirements.

a. Bottom down.
b. Top down.
c. Side down.
d. Bottom down at 45° angle.
d. Top down at 45° angle.

4.5.8.5 Ten foot drop test. The bare simulator shall be dropped from a height of 10 feet onto a 3 inch thick steel plate embedded in concrete. During First Article Testing, thirty items will be tested. Six items will be dropped for each of the five orientations listed below.

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a. Bottom down.
b. Top down.
c. Side down.
d. Bottom down at 45° angle.
e. Top down at 45° angle.

Observations shall be made for compliance with the applicable requirements. (Destructive test)

4.5.8.6 Seven foot drop test. The packed simulators shall be dropped from a height of 7 feet (measured to the bottom of the package) onto a 3 inch thick steel plate embedded in concrete in the orientations below after appropriate temperature conditioning:

a. One drop on side and one drop on end; at hot temperature.
b. One drop on top and one on bottom edge at 45° angle; at hot temperature.
c. One drop on bottom and one drop on top edge at 45° angle; at cold temperature.
d. One drop on side (opposite (a) side) and one drop on one end (opposite (a) end); at cold temperature.

The packed simulators shall be observed for compliance with the applicable requirements. (Non-Destructive test)

4.5.8.7 Forty-foot drop test. This test shall be conducted in accordance with Test 103 of MIL-STD-331. The packed simulators shall be dropped in the following positions after appropriate temperature conditioning:

a. One pack dropped with bottom down at cold temperature.
b. One pack dropped with end down at hot temperature.

Observations shall be made for compliance with the applicable requirements. (Destructive Test)

4.5.8.8 Loose cargo. This test shall be conducted in accordance with Method 514.3 of MIL-STD-810, Test Procedure II, Test Conditions I-3.2.3. Two boxes shall be positioned on end, one at hot and one at cold temperature, and two boxes shall be positioned on their base, one at hot and one at cold temperature. Observations shall be made for compliance with the applicable requirements. (Non-Destructive test)

4.5.8.9 Temperature humidity. The packed simulators shall be subjected to a 14 day temperature humidity test according to Test 105.1 of MIL-STD-331. Observations shall be made for compliance with the applicable requirements. (Non-Destructive test)

4.5.8.10 Temperature shock. The bare simulator shall be conditioned at the applicable temperature and cycled as specified in MIL-STD-810, Method 503.2, Table 503.2-1, hot dry induced temperature. Observations shall be made for compliance with the applicable requirements. (Non-Destructive test)

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4.5.8.11 Water immersion test. The bare simulator, without cap, shall be submerged in 12 inches minimum of water at ambient (70 + 5°F) temperature for a minimum of one hour. The simulator shall then be removed from the water, wiped dry, placed in an approved gunfire device, and functioned. Observations shall be made for compliance with the applicable requirements. (Non-Destructive Test)

4.5.9 Bridge wire resistance test. The igniter shall be tested for the required bridge-wire resistance with approved test equipment in accordance with 4.4.4. Observations shall be made for compliance with the applicable requirement. (Non-Destructive test)

4.5.10 Functional timing sequencing test. A continuity and open circuit test shall be performed on the electric igniter using an approved method and equipment in accordance with 4.4.4. Observation shall be made for compliance with applicable requirement. (Destructive Test)

4.5.11 Heat seal of barrier bag. (See Dwg. 9342970) The test shall be performed in accordance with the procedure specified in MIL-P-116 paragraph 4.4.4.3. After the test, the seal shall be examined for compliance with the applicable requirement (Destructive Test).

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Location of Celotex Panels for

M21 Simulator Fragmentation Test

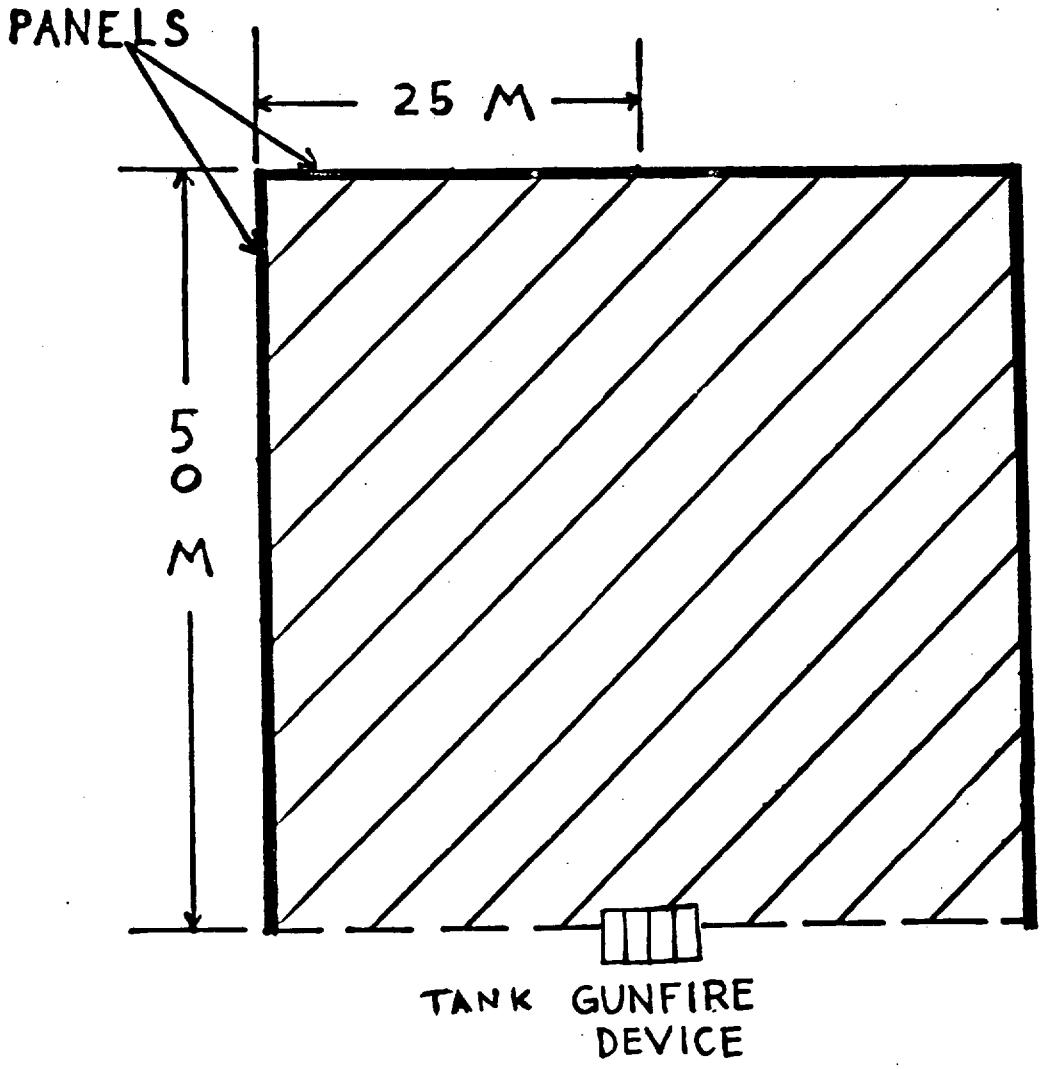


Figure 2

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4.5.12 Surface danger zone. Place insulation-wallboard panels (Celotex) 50 meters in front of and 25 meters (90 degrees) to each side of the gunfire device loaded with simulators as illustrated in Figure 2. The gunfire device shall be placed 3 ± .2 meters off the ground in a horizontal position facing the arena. Function all nine (9) rounds, one at a time. Examine Celotex panels for fragments after each round marking fragments, if any. If no fragments are collected on the panels, the surface danger zone is considered acceptable. If fragments are observed in the Celotex, remove, recording fragment weight, penetration depth and mean presented area. Determine the threshold velocity (V_{50}) using:

$$V_{50} = 493.52 \left[\frac{A}{W}\right]^{1/2}$$

Determine the striking velocity (V) using:

$$V = 1179\ W^{-.2899} Y^{.8101}$$

where: V_{50} = threshold fragment velocity (meters/sec)
V = striking fragment velocity (meters/sec)
Y = penetration depth (inches)
A = fragment mean presented area (sq cm)
W = fragment weight (grains)

If the striking velocity is lower than the threshold velocity, the fragment is considered non-hazardous and the surface danger zone is acceptable. If the striking velocity is higher than the threshold velocity, the fragment is considered hazardous and the danger zone is not acceptable. All data shall be submitted to ARDEC for evaluation and final disposition per 6.2.1.

NOTE: Celotex is the brand name for insulation board, thermal, interior, factory finish, 1/2 inch thick, NSN #5640 00 753 2803, with a density of 16.2 to 18.0 lb/ft^3. This or an equivalent commercial insulation board may be used. Boards should be stacked to a thickness of 1 inch minimum.

4.5.13 Connecting wire assembly resistance. The lead wire and bridge wire assembly shall be tested for the required resistance with approved test equipment in accordance with 4.4.4. Observations shall be made for compliance with applicable requirements. (Non-Destructive test)

4.5.14 Electromagnetic radiation susceptibility. The simulator, bare, packaged and in its gunfire device shall be tested in accordance with TOP 1-2-511, Appendix A. Observations shall be made for compliance with the applicable requirements. (Non-Destructive test)

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4.5.15 Test reports. Contractor performed test reports shall be reported in accordance with data item description DI-R-1724 (see 6.4) as specified in the contract. Government proving ground test results shall be documented and distributed as specified herein (see 6.2.1).

5. PACKAGING

5.1 Cleaning. The simulator shall be cleaned in accordance with Process C-1 of MIL-P-116, as required.

5.2 Drying. The simulator shall be dried in accordance with MIL-P-116, as required.

5.3 Preservation. Not required.

5.4 Packaging and marking level. Packaging and marking shall be as specified on drawing 9342970. If the dimensions of the simulator change, additional fillers shall be added to insure a tight fit or the dimensions of the fiberboard box and wood box changed accordingly.

5.5 Packing and marking level. Packing and marking shall be as specified on drawing 9342972.

6. NOTES

6.1 Intended use. The simulator is a device utilizing energy produced by explosives, propellants, or pyrotechnics or a combination thereof to produce a flash, bang and smoke signature. This is a training device used with the Hoffmann Device, EC Corp. Device or Joanell Laboratories Device.

6.2 Acquisition requirements. See contract.

6.2.1 Submission of test data and ammunition data cards. In addition to normal distribution of records, one copy of all test data and ammunition data cards shall be forwarded to each of the following: Commander, U.S. Army ARDEC, ATTN: AMSMC-QAT-M (D), and SMCAR-AEE-P, Picatinny Arsenal, New Jersey 07806-5000.

6.3 Submission of inspection equipment for design approval. See MIL-A-48078. The contractor shall submit inspection equipment designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.4, 4.5.12, 4.5.15	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.4.4	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-1714
6.2.2	Ammunition Data Cards	DI-MISC-80043

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L. Contractors in connection with specific acquisition functions should obtain copies from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4.1 Warranted ammunition. When the Quality Assurance Contract/Section E contains requirement for a performance warranty, the ammunition data card, DD1650, shall include the following warranty information:

Warranted ammunition: The performance of this lot of ammunition is under warranty until _____*_____.

* The warranty expiration date for each lot shall be inserted by the contractor.

6.5 Contractor's design requirements and specifications.

6.5.1 Inert and explosive materials. All inert and explosive materials used in the simulator shall be referenced to approved Government specifications if such exist. Proprietary materials should not be used unless specifically approved in writing by the contracting officer. At the direction of the contracting officer, the contractor should submit simulator drawings, in sufficient detail as to parts, special processes, and techniques to permit the preparation of documentation in accordance with 6.5.2.

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6.5.2 Drawings. Drawings should be in accordance with DOD-D-1000.

6.5.3 Interface drawings. Application for copies of EC Corp Device drawings should be addressed to: Project Manager for Training Devices, ATTN: AMCPM-TND-SLR, Naval Training Center, Orlando, Florida 32813-7100.

(Application for copies of Joanell Device drawings should be made to Armament Research, Development and Engineering Center, ATTN: SMCAR-AEE-P, Picatinny Arsenal, NJ 07806-5000. Request for a Hoffman Device for test purposes may be made to the Contracting Officer, Armament Munitions and Chemical Command, Rock Island, IL 61299.)

6.6 Noise protection. Noise protection is advisable to prevent ear damage whenever safe distance of 96 meters is not adhered to.

6.7 Subject term (key word) listing.
Simulator, M21
Flash
Sound
Hazardous
Gunfire
Training

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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SIMULATOR, ARTILLERY FLASH, M21
PERFORMANCE SPECIFICATION FOR

10 SCOPE:

10.1 Scope. This appendix details the quality assurance test methods required for testing the flash composition of the Artillery Flash Simulator, M21. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20 APPLICABLE DOCUMENTS.

20.1 Government documents.

20.1.1 Specifications, standards and handbooks. The following documents form a part of this appendix to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-1234	- Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1751	- Safety and Performance Test for Qualification of Explosives

20.1.2 Other Governments documents, drawings and publications. The following documents form a part of this appendix to the extent specified herein.

OD-44811	- Vol IV. Joint Safety and Performance Manual For Qualification of Explosives For Military Use

20.2 Non-Government publications. The following documents form a part of this appendix to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E-967	- Temperature Calibration of Differential Scanning Calorimeters and Differential Thermal Analyzers

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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30. REQUIREMENTS. This section is not applicable to this appendix.

40. TEST METHODS.

40.1 Impact test (ERL Type 12 tools).

40.1.1 Apparatus. The apparatus shall include the following:

a. ERL machine (Bruceton Machine)

(1) Striker pin, 1.250 inch diameter by 3.500 inches long, made of tool steel, hardened to 60-63 Rockwell "C" scale. Flat surface next to explosive is ground to a finish of 16 microinches.

(2) Flint sandpaper, 5/0 grade.

(3) Anvil, 1.250 inch diameter by 11.250 inches long, made of tool steel, hardened to 60 Rockwell "C" scale. All surfaces ground and polished, mounted in a tool holder assembly which is bolted to the machine base.

b. Microphone, Astatic Model JT-30C, ceramic type.

c. Thyratron tube, Model 2050.

d. Burlington Model 431 millivolt meter.

e. Weights, 2, 2.5 and 5 kilograms (weight commonly used is 2.5 kilograms), made of hardened steel. An electromagnet retains the drop weight until released by the operator.

40.1.2 Procedure.

40.1.2.1 Preparation of apparatus.

a. Figure 3 shows the assembly of the test elements to be used, which consists of a free falling weight, tooling to hold the energetic sample, and a supporting frame.

b. The microphone is mounted in the horizontal plane of the anvil face at a distance of 34 inches from the center of the anvil.

c. Calibrate the millivolt meter across a 100 ohm resistor by adjusting the setting to 50 millivolts.

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d. Adjust the amplifier gain to read 25 millivolts initially. Determine final adjustment by means of two test switches which make the thyratron tube alternately conductive and nonconductive. When proper gain setting has been achieved, the neon lamp will glow every time the thyratron is energized as demonstrated by 10 or more consecutive tests.

e. Calibrate the apparatus at least once each week for proper elimination of background noise. Adjust the instrumentation as described in paragraph d above. Release the drop weight from maximum height to impact on the test anvil. Under these conditions the neon lamp must not glow.

40.1.2.2 Preparation of sample. The test energetics are solid, granular materials which are either pure compounds or mixtures.

a. For materials which are normally cast-loaded into a weapon, prepare by casting as a thin sheet (weight from 3 to 10 grams depending on material availability and number of determinations to be made).

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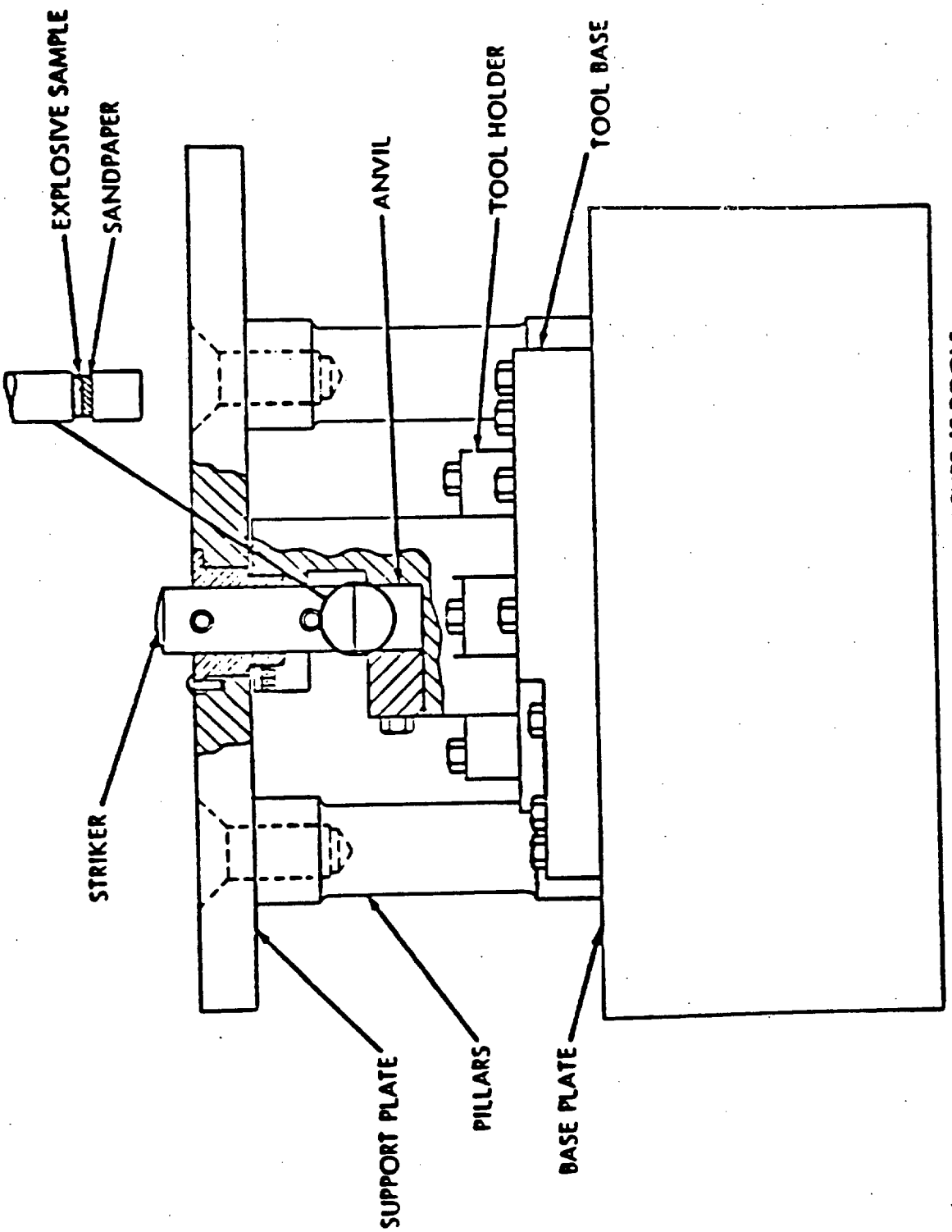


FIG. 3 TOOL HOLDER ASSEMBLY TYPE 12 TOOLS

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(1) Grind the cast sheet gently by hand in a wooden mortar and screen the material through a set of No. 16, 30 and 50 U.S. Standard Sieves. Carefully blend equal weights of material retained on the No. 30 and No. 50 sieves on a Fisher-Kendall mixer (simultaneous tumbling and stirring action) to furnish the test samples.

(2) Place each test sample of 35 ± 2 milligrams of energetic material in a loose pile in the center of the sandpaper.

b. Test other solid, granular materials by placing each test sample of 35 ± 2 milligrams of energetic material "as received" in a loose pile in the center of the sandpaper without further pretest processing.

40.1.2.3 Test procedure.

a. Place the test sample in the center of the anvil and lower the striker gently so that it rests on the top of the energetic pile.

b. Elevate the drop weight to a preselected height. Selection of the height used for the first drop is a matter of judgement. If the sensitivity of the test material has been previously measured, the first drop height will be chosen in the range where "fires" have occurred. If the material is of completely unknown sensitivity, an arbitrary starting height is used based on the sensitivity of similar compositions or the sensitivity which would be predicted from molecular structure.

c. Drop the weight and observe the result indicated by the noisemeter. If the neon lamp glows it is a "fire"; if not, the test is a "no-fire". The weight is caught by a sliding stop moved into position by the operator after initial rebound from contact with the striker. This prevents multiple impacts between weight and striker.

d. After the first fire is obtained (which may take 3 or 4 preliminary drops with an unknown material) successive drop heights are governed by the results of the previous drop according to the following procedure. Drop the weight from a height lower than the previous one by 0.093 log unit (where the log of a 10 centimeter drop is taken as 1.0). If the result is a fire, the next drop is 0.093 log unit lower; if a no-fire the next drop is 0.093 log unit higher. Continue testing by this "up and down" procedure for a total of 25 drops (usually called a "run").

e. After each drop, discard the test sample and use a fresh sample for the next drop. Clean the striker and anvil faces which are in contact with the test sample with solvent (such as acetone) after each test.

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f. Replace striker and anvil when working surfaces become roughened as determined by making carbon paper impressions of the surfaces. Old tools are refinished and reused. Replace the striker when its height has diminished by 0.250 inches.

40.1.3 Calculations.

a. For each test, record the log of the height from which the weight was dropped and the decision as to whether the drop resulted in a fire or no-fire.

b. Examine the data to determine whether more fires or no-fires occurred (if the numbers are equal, either may be used). Select the lower number for analysis and discard the balance of the data. Summarize statistically by the use of the following table (numbers are for illustration only):

log	i	n_i	i x n_i	i^2 x n_i
1.7	0	2	0	0
1.8	1	10	10	10
1.9	2	10	20	40

c. The log of a given drop height is entered in the first column. They are arranged in ascending order, starting with the lowest for which a test is recorded as indicated in the example above. In the next column, "i" is a consecutive number corresponding to the number of equal increments above the base, or "zero" line. The next column, "n_i", tabulates the number of fires (or no-fires) which occurred at i=0, i=1, i=2, etc. The other columns are computations of i times n_i and i^2 times n_i.

d. Compute the mean from the formula:

$$m = c + d\left(\frac{A}{N} \pm \frac{1}{2}\right)$$

Where:
$N = \Sigma n_i$
$A = \Sigma i \times n_i$
c = normalized height of the lowest line (i_0)
d = normalized interval between drops (0.093)

The sign inside the parenthesis is (+) if no-fires and (-) if fires are used.

e. Report the mean computed as the "50% point". It represents a 50% probability of fire.

f. The number may be reported in log units as determined directly from the computation. More often, the antilog is found and this is reported as a height in centimeters. Often, the standard deviation is also estimated by the following technique:

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A number, "M", is computed from the formula

$$M = \frac{\sum i^2 n_i}{N} - \left[\frac{\sum i n_i}{N}\right]^2$$

Using a table or graph appearing in reference (2), a value "s" is obtained. The standard deviation (σ) is then:

$$\sigma = ds$$

It is always expressed in log units.

g. The table below sets out typical test results for eight common energetics.

	50% Point	
Energetic	cm	σ
Lead azide	4	0.12
PETN	12	0.13
RDX	24	0.11
HMX	26	0.10
Tetryl	38	0.07
Comp B	60	0.13
TNT	157	0.10
Explosive D	254	0.05

h. Reference for these calculations are as follows:

(1) Dixon, W.J. and Massey, F.J., Jr., "Introduction to Statistical Analysis", McGraw-Hill, 1951, pp 278-286.

(2) Applied Mathematics Panel of the National Defense Research Committee (AMP Report No. 101.1R SRG-P No. 40).

40.2 Friction pendulum. This test shall be conducted in accordance with Friction Sensitivity Method (page 5.7) as found in Vol. IV Joint Service Safety and Performance Manual for Qualification of Explosives for Military use.

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40.3 Electrostatic discharge (0.25 joule). This test shall be conducted in accordance with METHOD 4 of MIL-STD-1751.

40.4 Explosive temperature (5 seconds).

40.4.1 Apparatus. The apparatus for this test includes the following:

a. A barricaded press with a die for loading 40 milligrams of the energetic sample into a No. 8 blasting cap.

b. A crimper.

c. A furnace with a molten Woods metal bath and shield.

d. Associated temperature measurement and controls.

e. A time interval recorder.

40.4.2 Procedure.

40.4.2.1 Preparation of apparatus.

a. Check press operation and set loading force for 230 pounds with Dillion gauge.

b. Inspect equipment and empty blasting cap for cleanliness.

c. Place an empty blasting cap in the loading fixture.

d. Weigh out the proper amount of energetic material (40 mg), or use a previously calibrated conductive volumetric scoop, and pour into cap using a loading funnel.

e. Insert a copper gas check, open side up, into cap, and manually push the gas check down to bottom of cap with rod.

f. Put 30 mg of dried talc in cap, insert second gas check, and repeat step e.

g. Close barricade door and apply 6000 psi pressure (230 pounds loading force) for ten seconds.

h. Remove die from press. Behind a shield, remove cap from die with shielded tweezers and transfer cap to crimper.

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i. Push cap into crimper with shielded tweezers and hold against stop.

j. Slowly lower handle of crimper until resistance is felt, then remove tweezers and crimp until stop is reached.

k. Raise handle of crimper and remove loaded cap with tweezers and place in nonpropagating tray behind barricade.

40.4.2.2 Test procedure.

a. Put a piece of masking tape around each cap.

b. Heat Woods metal bath to highest desired temperature. Stir bath with steel rod and wait at least five minutes until temperature is uniform.

c. Rotate sample holder to loading position.

d. Reset timer to zero. Check start and stop signals, then reset zero.

e. Insert loaded cap in hole in fixture using shielded tweezers, and press into position. (Tape should hold cap firmly up, if not, add or remove tape as necessary).

f. Grasp handle and rotate fixture to allow sample to drop into bath.

g. When an explosion occurs, record time and temperature, and using handle, lift fixture and rotate back to the loading position.

h. If an explosion does not occur within two minutes, remove sample from the bath as described in step g and allow the sample to cool for at least one minute. Then destroy the sample; using shielded tweezers, place the cap behind a hood safety shield in a hole in a shielded steel block on a hot plate or in a furnace at 400°C to destroy the energetic material at the end of the test, allow sample to cool and then dispose of in an explosive waste can (underwater).

i. After all samples to be fired at the temperature of the bath have been fired, lower the temperature setting on the controller to the next temperature desired. Add cold metal, then stir thoroughly and check temperature. If still too hot, add more cold metal and stir again. Remove excess metal with a spoon and pour onto a steel plate to cool for future use. As long as the temperature is known at the time the sample is tested, the exact temperature set on the controller is not required.

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j. Perform successive tests in the same manner to obtain time-to-explosion temperature values over a time range of 0.5 to 9 seconds.

40.4.2.3 Calculations.

a. Report the test results as the temperature in degrees centigrade which corresponds to the 5-second explosion temperature.

b. Determine the 5-second explosion temperature as follows. Plot the data on semilog paper, the logarithm of the time-to-explosion versus the reciprocal of the absolute temperature. This results in a straight line. A linear least square curve fit program is used to determine the temperature value for the 5 second time-to-explosion.

40.5 Vacuum thermal compatability (stability 100°C). This test shall be conducted in accordance with Method 503.1.1 of MIL-STD-1234.

40.6 Auto ignition temperature.

40.6.1 Apparatus.

a. Deltatherm II Differential Thermal Analysis Accessory or equivalent.

b. Mosely 7100B two-pen time-base recorder or equivalent.

40.6.2 Procedure.

a. Use a sample size which will produce an isosceles triangle of not too great an angle. Too steep an angle (80-90°) is not acceptable (too large a sample); and too gradual an angle (0-35°) is not acceptable (too small a sample). Start with .2 to .5 milligrams for 10°/minute and 2 milligrams for 2.5°/minute.

b. Follow the general procedure for Differential Thermal Analysis, ASTM E967-83.

c. Place the energetic sample and the reference material into aluminum or steel cups (with size of up to 4.8 millimeters) and place into the thermo cups (steel cups to which an alumel and a chrome wire are welded).

d. Close experimental area and set heating rate. The heating rates to be used shall be: 10, 5, 2 1/2, and 1 degrees centigrade per minute in each of four separate tests.

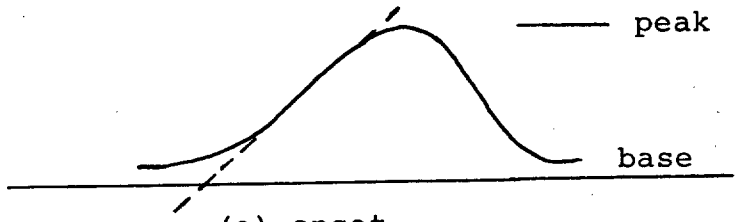
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e. Set limits on recorder (paper speed, sensitivity) and instrument sensitivity. One pen shall record the temperature of the reference material; the other pen shall record the ΔT of the 2 cups at a space of 1/10 of an inch ahead of the temperature pen.

f. To minimize the autocatalytic activity, pass a constant flow of argon across the sample holder while heating the sample at a constant rate.

40.6.3 Calculations.

a. Determine extrapolated onset(e) and peak temperature by using the following guidelines.



(e) onset

b. Graph ϕ/T^2 vs. T^{-1}, where ϕ = heating rate, $^{o}C/min$, T = peak temperature, to obtain the activation energy (slope).

c. The following relationship is used to solve for the necessary variables.

$$E\phi/RT^2 = Ae^{-E/RT} = 0.05^{o}C/min$$

d. Obtain A, the preexponential factor, at each peak temperature (at the four heating rates) and use the average of these values for the equation. This is done by solving for A in the above equation. (See Table I excerpt from Thermochimica Acta, 14 (1976), 183-199).

e. Set $E\phi/RT^2$ equal to 0.05 and solve for the autoignition temperature from T in the exponent.

50. PACKAGING. This section is not applicable to this appendix.

60. NOTES. This section is not applicable to this appendix.

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TABLE I. EXCERPT FROM THERMOCHIMICA ACTA

CALCULATION OF ACTIVATION ENERGY AND AUTOIGNITION TEMPERATURE FOR TNT (CONFINED) KNK 11-270

Temperature (°C)	Temperature (K)	Rate = ϕ (°C min^{-1})	$1/T \times 10^3$ (K)	$\phi/T^2 \times 10^{-5}$
Temperature at the peak of the exotherm:				
333	606	22.0	1.650	5.847
315	588	10.7	1.701	3.058
296	569	5.3	1.757	1.637
287	560	2.6	1.786	0.829
Temperature at the beginning of the slope to the exotherm peak:				
326	599	22.0	1.666	5.988
309	582	10.7	1.718	3.121
292	565	5.3	1.770	1.660
285	558	2.6	1.792	0.835

$$\text{Slope} = \frac{E}{\text{Ln } R} = \frac{\log(\phi/T^2)_1 - (\phi/T^2)_2}{1/T_1 - 1/T_2}$$

E_p = 24200 cal mol^{-1}; E_s = 24900 cal mol^{-1}.
E_p = activation energy at peak; E_s = activation energy at slope;
ϕ = constant heating rate; T = temperature (K); R = gas constant;
A = frequency factor (sec^{-1}).

Peak temp. (K)	$\phi/T^2 \times 10^{-5}$	$\frac{E\phi}{RT^2}$	$e^{-E/RT} \times 10^{-9}$	A x 10^8
606	5.991	0.72965	1.865	3.912
588	3.095	0.37695	1.013	3.721
568	1.642	0.1998	0.488	4.094
560	0.829	0.10096	0.358	(2.821)
			average (3)	= 3.90 x 10^8
Slope temp. (K)				
599	6.131	0.76799	0.830	9.253
582	3.159	0.39571	0.451	8.774
565	1.660	0.20794	0.235	8.849
558	0.834	0.10447	0.178	(5.869)
			average (3)	= 8.96 x 10^8

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$$\frac{E\phi}{RT^2} = Ae^{-E/RT}$$

as $\phi = 0$

$$E\phi/RT^2 \qquad 0.05$$

$$\log 0.1 = \log 3.90 \times 10^8 - \frac{E}{RT \times 2.303}$$

T = 552K
279.0°C peak temperature

$$\log 0.05 = \log 8.96 \times 10^8 - \frac{E}{RT \times 2.303}$$

T = 547.6K
= 274.6°C autoignition temperature confined.

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Custodian:
Army-AR

Preparing activity:
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